# Number 5 — Calculators · Estimating · Upper and lower bounds

Calculator allowed except in question 4. Marks in brackets. ★ = harder question.

**1** Calculate, giving each answer to 3 s.f.: [4]

**a** $2.4 + 2.4^2 + 2.4^3$ **b** $\dfrac{2.6 \times 10^3}{2.6^3}$ **c** $\dfrac{\sqrt{230}}{5.1 \times 10^3 - 3.8 \times 10^2}$

**d** $\dfrac{\sqrt{8.1 \times 10^3}}{\sqrt{2.5 \times 10^{-5}}}$

**a** $2.4 + 5.76 + 13.824 = 21.984 \approx 22.0$. **b** $2600 \div 17.576 = 148$. **c** Brackets round the denominator: $15.166 \div 4720 = 0.00321 = 3.21 \times 10^{-3}$. **d** $90 \div 0.005 = 18\,000 = 1.80 \times 10^4$.

**2** Write the lower and upper bounds of each measurement. [8]

**a** 34 m (nearest metre) **b** 62 kg (nearest kg) **c** 480 s (nearest 10 s)

**d** 2300 km (nearest 100 km)

Half a unit either side. **a** $33.5 \le x < 34.5$ **b** $61.5 \le x < 62.5$ **c** $475 \le x < 485$ **d** $2250 \le x < 2350$.

**3** $w = \dfrac{x + y}{z}$ where $x = 3.5 \pm 0.5$, $y = 2 \pm 0.5$ and $z = 0.8 \pm 0.1$. Find the lower and upper bounds of $w$. [4]

Largest $w$: biggest top, smallest bottom: $\dfrac{4 + 2.5}{0.7} = 9.29$. Smallest $w$: $\dfrac{3 + 1.5}{0.9} = 5$. So $5 \le w < 9.29$ (3 s.f.).

**4** Without a calculator, estimate each value in standard form to 1 s.f. Show your rounding. [4]

**a** $2.1 \times 10^3 \times 3.9 \times 10^2$ **b** $\dfrac{6.1 \times 10^4}{2.9 \times 10^2}$ **c** $\sqrt{8.9 \times 10^4}$ **d** $\dfrac{5.1\pi \times 10^3}{\sqrt{9.3}}$

Round everything to 1 s.f. first. **a** $2 \times 4 \times 10^5 = 8 \times 10^5$. **b** $\dfrac{6}{3} \times 10^2 = 2 \times 10^2$. **c** $\sqrt{9 \times 10^4} = 3 \times 10^2$. **d** $\dfrac{5 \times 3 \times 10^3}{3} = 5 \times 10^3$.

**5** A runner covers 400 m, measured to the nearest 10 m, in 52 s, measured to the nearest second. Find the lower and upper bounds of her average speed in m/s. [3]

Fastest: longest distance over shortest time: $\dfrac{405}{51.5} = 7.86$ m/s. Slowest: $\dfrac{395}{52.5} = 7.52$ m/s. So $7.52 \le v < 7.86$ m/s.

**6** The area of a semicircle is measured as 60 m² to the nearest 5 m². Find the lower and upper bounds of its perimeter. (Area of a circle $\pi r^2$, circumference $2\pi r$.) [2]

> Area bounds $57.5 \le A < 62.5$. For a semicircle $A = \frac{1}{2}\pi r^2 \Rightarrow r = \sqrt{\dfrac{2A}{\pi}}$: $r$ from $\sqrt{115/\pi} = 6.05$ to $\sqrt{125/\pi} = 6.31$ m. Perimeter $= \pi r + 2r = r(\pi + 2)$: from $31.1$ m to $32.4$ m.

**Total: 25 marks**

Check: bounds are ± half the unit of rounding, lower bound included, upper not · for a quotient, max = max ÷ min and min = min ÷ max · estimate by rounding to 1 s.f. before calculating · brackets round numerators and denominators on the calculator.